

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

Part I

Theory

December 6, 1975

- Instructions:
1. Open Book and Open Notes.
 2. Do four of the following six problems.
 3. Use a separate blue book for this section of the exam.
 4. Indicate which questions you wish graded.
-

1. Suppose X is an observation from a population with density given by

$$f_{\theta}(x) = \frac{1}{\theta} e^{-x/\theta} \quad x > 0 \text{ for } \theta > 0.$$

Give (explicitly) a formula for a $100(1-\alpha)\%$ confidence interval for θ .

2. Suppose X has a density of the form given in problem 1. It is desired to test the hypothesis $H_0: \theta \leq 1$ vs. $H_1: \theta > 1$.

- a) Find a test of size $\alpha = e^{-3}$ for the above hypothesis.
- b) Is the test you found in part (a) uniformly most powerful? If so, why? If not, why not?

3. Suppose $X_1, X_2, \dots, X_n$ are independent observations on a population with density given in problem 1.

- a) Find the M.L.E. of $1/\theta$.
- b) Discuss whether there exists an unbiased estimator of $1/\theta$ or not. If such exists find the best one you can and tell in what sense it is best. (Hint: The answer may depend on n .)

4. Let $X_1, X_2, \dots, X_n$ be independent observations on a population with density ($\theta > 0$)

$$f_{\theta}(x) = \frac{\theta^{k+1}}{k!} x^k e^{-x\theta}, \quad x > 0$$

If loss is measured by squared error and the prior distribution on θ has

$$\text{density } g(\theta) = \frac{\theta^j e^{-\theta}}{j!}, \quad \theta > 0,$$

- a) Find the Bayes estimator of θ .
- b) Is the Bayes estimator unbiased? Show why or why not.

5. If the probability that a family will have exactly n children is 2^{-n} for $n=1,2,\dots$ and if all 2^n permutations of the sexes of the n children are equally likely, what is the probability that a family will have no boys? The following formula will be helpful:

$$\sum_{n=0}^{\infty} X^n = \frac{1}{1-X}, \quad -1 < X < 1.$$

6. In a random sample of size K drawn without replacement from an urn containing r red balls, b black balls and w white balls, derive the conditional probability that the sample will contain exactly i white balls given that it contains exactly j red balls.

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Part II

Applied

December 6, 1975

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1. A field experiment was conducted to estimate the effects of eight (8) fertilizers on the yield of oats. Instead of harvesting each plot separately six samples were harvested from each plot. The ANOVA conducted on the data was

	<u>Source</u>	<u>df</u>	<u>ms</u>
	Total	239	
(F)	Treatments	7	640
(R)	Replicates	4	320
(R)	Experimental Error		160
(R)	Sampling Error		80

- (a) Fill in the missing df appropriate to the design and write out the ems.
 - (b) Define more specifically what the two sources of "Error" are in this particular experiment.
 - (c) Test the hypothesis that the population mean yields for the eight treatments are all the same. Interpret your results verbally.
 - (d) Assuming that the mean yield for one of the treatments is 40 bushels per acre, compute and interpret verbally its 95% confidence interval.
 - (e) Estimate the variance of a treatment mean if six (6) replicates and four (4) samples per plot were to be used. What would be the gain or loss in information compared to the use of five (5) replicates and six (6) samples per plot?
2. Someone claimed that a person could not distinguish between two foods eaten when he could not smell, especially when the foods were about alike in texture and flavor. An expert selected two foods, A and B, as being types which would qualify under the claim if it were true. A test was arranged as follows:

Each of 100 persons was given four portions at random to taste; three of one type and one of the other. All wore nose clips during the test. Each was asked to pick out the "odd" portion.

- (a) State the null and alternative hypotheses.
- (b) Test the claim at the 5% level if 30 of the 100 people correctly identified the "odd" portion.
- (c) What conclusion can be drawn from this experiment?

3. A new variety of tomato has been developed. It is desired to know if the yield of this variety is greater than the yield of a standard variety. An experiment is conducted and the yield was recorded. The following is a summary of the results of the experiment.

<u>New Variety</u>	<u>Standard Variety</u>
$n = 8$	$n = 16$
$\bar{x} = 25.3$	$\bar{x} = 22.3$
$s^2 = 9$	$s^2 = 25$

- (a) Determine if there is sufficient evidence to indicate that the yield of this new variety is larger than that of the old variety. ($\alpha = .05$).
- (b) Obtain a 90% confidence interval for the difference between the mean yields of the two varieties of tomato.
4. It is believed that the relationship between the % of catalyst in a chemical process and the yield is linear. From the data given below
- (a) Fit the least square regression line.
- (b) Test for linearity.
- (c) Describe how one would proceed if the linear model is inadequate.

<u>% Catalyst = x</u>	<u>Yield = y</u>
4	70
5	71
6	73
7	74
7	73
8	76
8	75
9	77
10	79
11	81
$\Sigma x = 75$	$\Sigma y = 749$
$\Sigma xy = 5684$	$\Sigma y^2 = 56207$
$\Sigma x^2 = 605$	

5. An experiment was conducted to compare the effect of two diets on the body weight gain of rats. Four rats were given Diet 1 and another four rats were given Diet 2. Their body weight gains are shown below.

<u>Diet 1</u>	<u>Diet 2</u>
0	0
1	-1
2	-2
3	-3

- (a) Use a t-test to compare the diet means. (Assume $\sigma_1^2 = \sigma_2^2$)
- (b) Use an Analysis of Variance to compare the diet means.
- (c) Use the least significant difference procedure to compare the diet means.
- (d) Do the results obtained from the three procedures above agree? Is this surprising? Carefully explain why.
6. To study the performance of a newly designed motorboat it was timed over a marked course under various wind and water conditions. The following data (in minutes) was obtained.

	<u>Conditions</u>		
<u>Calm</u>	<u>Moderate</u>	<u>Choppy</u>	
20	21	26	
17	23	24	
14	16	23	
<u>24</u>	25	29	
75	18	<u>21</u>	
	<u>23</u>	123	
	126		

- (a) Complete the following ANOVA Table.

<u>Source</u>	<u>df</u>	<u>ss</u>	<u>ms</u>	<u>F</u>
Total				
Treatment		79.65		3.25
Error				

- (b) Test the hypothesis that the boat's performance is not affected by the differences in wind and water conditions.
- (c) Complete the analysis of this data including the use of some multiple comparison procedure.

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Part I

Theory

March 15, 1975

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1. Someone rolls a pair of dice and tells you the result is odd and exceeds 6. What is the conditional probability that it is a 7 or 11?
2. Let X and Y be random variables with a joint density given by

$$f(x,y) = \begin{cases} 2 & 0 < x < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find (a) The marginal density of Y

and (b) $P[Y < 2X]$

(c) Are X and Y independent? Prove or Disprove.

3. Given a sample of size n from a Poisson population with unknown mean λ , find the minimum variance unbiased estimator of $\lambda^2 + 3\lambda$.
4. Let X be a single observation on a population with density

$$f_{\theta}(x) = \begin{cases} \frac{3x^2}{\theta^3} & 0 < x < \theta \\ 0 & \text{otherwise} \end{cases}$$

Find a $100(1-\alpha)\%$ confidence interval for θ .

5. Let $X_1, \dots, X_n$ be a random sample from a population with density given by

$$f_{\lambda}(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the M.L.E. of λ .

(b) Find the Bayes estimator of λ if loss is squared error and the prior has density $g(\lambda) = e^{-\lambda}$, $\lambda > 0$.

6. Let $f_0(x) = \frac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}$ and $f_1(x) = \frac{e^{-\frac{1}{2}(x-1)^2}}{\sqrt{2\pi}}$.

It is desired to test $H_0: f=f_0$ vs $H_1: f=f_1$. A single observation X is taken and a critical region $0 < x < 1$ is selected. For this critical region,

- (a) Find α .
- (b) Find β , the power.
- (c) Is this the best critical region of its size? If not what region is best?

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- (c) Is this the best critical region of its size? If not what region is best?

RUTGERS-THE STATE UNIVERSITY
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Part II

Applied

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1. A biologist is seeking a way of distinguishing between two organisms A and B. He decides to seek a growing medium in which the two organisms would have sharply different growing rates. He runs the following experiment.

		<u>Nutrient</u>			
		1	2	3	
		Organism		Organism	
		A	B	A	B
Day 1		3	4	1	6
2		6	5	2	8
				10	8

where the response (y) is the volume of growth

Nutrient and organism may be treated as fixed variables. He is only able to prepare 6 agar dishes per day, so he decides to include day 1 and 2 as a block effect.

Some calculations

$$\Sigma y^2 = 440$$

$$\Sigma y = 66$$

$$(\Sigma y)^2 / n = 363$$

$$9^2 + 9^2 + 3^2 + 14^2 + 17^2 + 14^2 = 852$$

$$TN_1 = 18 \quad TO_A = 29 \quad TB_1 = 27$$

$$TN_2 = 17 \quad TO_B = 37 \quad TB_2 = 39$$

$$TN_3 = 31$$

$$\frac{(18)^2 + (17)^2 + (31)^2}{4} = 393.5$$

$$\frac{(27)^2 + (39)^2}{6} = 375$$

$$\frac{(29)^2 + (37)^2}{6} = 368.3$$

- (a) Construct the ANOVA table including the E.M.S. column. Show calculations.
- (b) Relate the question asked by the biologist to the ANOVA table and to the original data.
- (c) Does he have a method?
- (d) Calculate a confidence interval for organism A.
2. In a triangular test for selecting judges to compose a taste, a prospective judge was successful in selecting the odd sample 11 times in 15 trials. Would you select him for the panel? How many would he have to pick correctly to be chosen? What is the probability of a Type I error if we accept the above judge for our panel?
3. Suppose you run a completely randomized experiment with five levels of one variable and obtain the following results.

	<u>Treatment</u>				
	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>	<u>E</u>
Average	15	20	18	30	32

Each treatment contained 4 observations. The F test rejects the assumption that the means constitute a homogeneous group.

The residual mean square from the ANOVA is 25.

- (a) Use two procedures to suggest alternative hypotheses.
- (b) Calculate a 95% confidence interval for

$$\left[\frac{\mu_A + \mu_B}{2} - \frac{\mu_D + \mu_E}{2} \right]$$

4. It is believed that the relationship between the % of catalyst in a chemical process and the yield is linear. From the data given below
- (a) Fit the least square regression line.
- (b) Test for linearity.
- (c) Describe how one would proceed if the linear model is inadequate.

<u>% Catalyst = x</u>	<u>Yield = y</u>
4	70
5	71
6	73
7	74
7	73
8	76
8	75
9	77
10	79
11	81
$\Sigma x = 75$	$\Sigma y = 749$
$\Sigma xy = 5684$	$\Sigma y^2 = 56207$
$\Sigma x^2 = 605$	

5. An experimenter wishes to compare six treatments. He has available possible sources of experimental units. For the same cost he has his choice of:
- (a) 36 units which come naturally classified into 6 rows and 6 columns and for which the anticipated residual variance is 10.
 - (b) 48 units which come naturally classified into 4 blocks of twelve units each and having an anticipated within-block variance of 14.
 - (c) 30 units which come naturally classified into 5 blocks of 6 units each with anticipated within-block variance of 8.
 - (d) 21 units which come naturally classified into 7 blocks of 3 units each with anticipated within-block variance of 3.

Assuming that interactions can be ignored

- (1) Give your recommended design for each of these.
- (2) Give your order of preference (with justification) based on statistical considerations only.

6. Measurements of the yields of two processes, A and B, have shown average yields of $\bar{X}_A = 46.5$, $\bar{X}_B = 47.1$. The process yields are independent normal variables with means μ_A and μ_B and variances, $\sigma_A^2 = 25$ and $\sigma_B^2 = 16$ respectively. $\bar{X}_A$ is based on 100 observations and $\bar{X}_B$ on 64. We want to test $H_0: \mu_A = \mu_B$ with a 2-sided, 10% level test.
- (a) Give the Critical Region of the test.
 - (b) Carry out the test for the data above.
 - (c) If $\mu_A = 50$ and $\mu_B = 48$, what is the power of this test?
 - (d) Give a 90% confidence interval for $(\mu_A - \mu_B)$.

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December 14, 1974

Theory

Part I

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1. Let $X_1, \dots, X_n$ be a sample of i.i.d. observations from a normal population with mean μ and known variance, 1. It is desired to estimate e^μ .

- (a) Find the M.L.E. of e^μ .
- (b) Is the M.L.E. unbiased?
- (c) Find the minimum variance unbiased estimate of e^μ .

2. Let X be a single observation from a population with density

$$f_\theta(x) = \begin{cases} \frac{2x}{\theta^2} & 0 \leq x \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

Give a $100(1-\alpha)\%$ confidence interval for θ .

(It may help you to note that if you observe $X=x$ you know that $\theta \geq x$. Hence you should use an interval that looks like $(x, ?)$.) Do you recognize the distribution?

3. A single observation X is taken from one of the following discrete distributions P .

		Values of X					
P_0		0	1	2	3	4	5
	$P[X=x]$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{3}{16}$	$\frac{5}{16}$

		Values of X					
P_1		1	2	3	4	5	6
	$P[X=x]$	$\frac{3}{32}$	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{5}{32}$	$\frac{4}{32}$	$\frac{17}{32}$

Find the most powerful test of $H_0: P=P_0$ vs $H_1: P=P_1$ of size $\alpha = 5/16$.

4. Suppose an extremely accurate marksman is shooting a projectile at a target of radius 1. If (θ, r) represents the polar coordinates of the hit, we assume the joint density of (θ, r) is given by

$$g(\theta, r) = \begin{cases} \frac{1}{2\pi} g(r) & 0 \leq r \leq 1 \\ & 0 \leq \theta \leq 2\pi \\ 0 & \text{otherwise} \end{cases}$$

(When $\int_0^1 g(r) dr = 1$). Let

$$A = \{(\theta, r) : -\pi \leq \theta \leq 0, 0 \leq r \leq 1\}$$

$$B = \{(\theta, r) : \pi/2 \leq \theta \leq 3\pi/2, 0 \leq r \leq 1\}$$

- (a) Show that the sets A and B are independent.
(b) Find a third set C such that $P(C) = \frac{1}{2}$ and A, B, and C are mutually independent.

5. Suppose the Tic Tac Toe Cereal Company has two production lines. One line produces X's and the other O's. The hourly production of X's and O's are independent Poisson variates with means $\mu = 1000$ and $\lambda = 600$ respectively. At the end of each hour the hourly production of X's and O's is put into a box and sold to an eager customer. If a certain box has a total of 1200 X's and O's, what is the approximate probability that there are at least 800 X's? (Hint: the conditional distribution of the number of X's given the total number of X's and O's is a well known discrete distribution beginning with a B _____. Find it, approximate it and go on to the next question).

6. Let $X_1, X_2, \dots, X_n$ be a sample from a Poisson distribution with mean λ . It is desired to estimate λ with loss $L(a, \lambda) = (a - \lambda)^2$. Let the prior density of λ be

$$g(\lambda) = \begin{cases} e^{-\lambda} & \lambda > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the Bayes estimator of λ .
- (b) Find the Bias and Variance of the above estimator.

RUTGERS-THE STATE UNIVERSITY
Master's Comprehensive Examination

December 14, 1974

Part II

Applied

- Instructions:
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1. Measurements of the yield of two processes, A and B, have shown average yields of $\bar{X}_A = 46.5$, $\bar{X}_B = 47.1$. The process yields are independent normal variables with means μ_A and μ_B and variances $\sigma_A^2 = 25$ and $\sigma_B^2 = 16$. $\bar{X}_A$ is based on 100 observations and $\bar{X}_B$ on 64. We want to test $H_0: \mu_A = \mu_B$ with a 2-sided, 10% level test.
- (a) Give the critical region of the test.
 - (b) Carry out the test for the data above.
 - (c) If $\mu_A = 50$ and $\mu_B = 48$, what is the power of this test?
 - (d) Give a 90% confidence interval for $(\mu_A - \mu_B)$.
2. (a) A researcher wants to study the effectiveness of flu vaccination. He has asked each of 90 people whether he had had flu during the previous winter and also whether he had been vaccinated. Of the 90 people sampled, 18 had been vaccinated, and 45 had had flu. Of the 18 vaccinated people, only 3 had had flu. Should the experimenter conclude that vaccination and contracting flu are independent events or not?
- (b) A model for genetic inheritance predicts that 3 recognizable types (A, B, and C) should exist in a 6:1:1 ratio in a population. To test this model, 48 individuals were sampled at random and each classified according to type.

TYPE			48
A	B	C	
30	6	12	

- Do the data suggest that the genetic model is a reasonable one for describing the inheritance of the trait being considered?
- (c) For the data in part (b) write a 95% confidence interval for the true proportion of type C in the population.
3. In parts (a), (b), and (c) write out the appropriate ANOVA table showing the sources of variation, the d.f., and the e.m.s. Also indicate in each case how the stated question or questions may be answered including any further analyses that you feel are appropriate.
- (a) A study was conducted to obtain information about the variability of income of farms in the midwest. Eight (8) types of farming areas were included in the study, and within each area five (5) counties were selected at random. Within each of the chosen counties twenty (20) farms were selected at random and the farm incomes were recorded.
- (b) An experiment was performed to assess the relative effects of five fertilizers on the yield of a certain variety of oats. The location of the thirty (30) experimental plots available for use in the experimentation was such that it seemed advisable to group the plots into six (6) blocks of five (5) plots each. The treatments were then randomly assigned to the plots within each block. At the end of the growing season each plot was harvested.

- (c) A process is designed to produce a fishline that will have a 15 lb. test rating. The line may be treated with four (4) different water proofings. The hypothesis is that the four treatments have the same, if any, effect on the test rating of the cord. Twenty (20) samples of each type of treated cord are tested for breaking strength.

4. A simple linear regression model

$$Y = \alpha + \beta X + \epsilon$$

was fit to a set of data. The following results were found.

$$b = 4$$

$$\bar{X} = 4$$

$$\bar{Y} = 20$$

Additionally part of the analysis of variance table is summarized below.

<u>Source</u>	<u>df</u>	<u>ss</u>	<u>ms</u>	<u>F</u>
Total	10	800		
Regression				
Residual		288		

- (a) Find the estimated regression line
- (b) If it is assumed that ϵ is distributed as $N(0, \sigma^2)$ then estimate σ^2 .
- (c) Test the hypothesis that $\beta=0$. Let the probability of a Type I error = .10.
- (d) Find a 95% confidence interval for the mean value of Y when $X=4$.
- (e) Find an estimate of the correlation coefficient.
- (f) Find the coefficient of determination, r^2 . What does this mean?
- (g) Test the hypothesis that $\rho=0$.

5. Below is an abbreviated ANOVA table.

<u>Source</u>	<u>df</u>	<u>ms</u>
Total	31	--
A	3	240
B	3	120
AxB	9	60
obs(AxB)	16	30

In each of the following situations obtain the e.m.s., indicate the appropriate null and alternative hypotheses concerning A, B, and AxB, and test each hypothesis.

- (a) A is random and B is fixed.
6. Suppose we have a 2^4 experiment with factors A, B, C, D and wish to obtain a $\frac{1}{2}$ replicate by using ABD as the defining contrast.
- (a) What treatments in addition to (1) are used? [note: (1) is Yate's notation representing the treatment combination for which A, B, C, and D are all at their low levels.]
- (b) How many experimental units are used?
- (c) What are the aliases of all main effects and two factor interactions?
- (d) Would you prefer another interaction in place of ABD in constructing a $\frac{1}{2}$ replicate? Why?

RUTGERS-~~THE~~ STATE UNIVERSITY
Master's Comprehensive Examination

March 16, 1974

Part I

Theory

Instructions: Open Book and Open Notes. Do four of the following six problems. Use a separate blue book for this section of the exam. Be sure to indicate which questions you wish graded.

1. X_1, X_2 are two independent observations from a uniform distribution in the interval $(0, \theta)$. Show that $(Y, \frac{Y}{\sqrt{\alpha}})$ is a confidence interval for θ with confidence coefficient $1 - \alpha$, where $Y = \max(X_1, X_2)$.

2. A fair coin is tossed until two heads are obtained. Show that the average number of tosses required is 4.

(Note: $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ for $|r| < 1$).

3. Someone rolls a pair of dice and tells you the result is odd and exceeds 6. What is the conditional probability that it is a 7 or 11?

4. Suppose n electron tubes are placed on life test. The experiment is stopped when all tubes have failed, the times of failure being $x_1, \dots, x_n$. Assume that the time-to-failure has an exponential distribution with probability density function given by

$$f(x; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}} \quad x \geq 0, \theta > 0$$

$$= 0 \quad x < 0$$

where the parameter θ is the mean of the distribution.

- (a) Derive the maximum likelihood estimator $\hat{\theta}$ of the mean life θ . Show work in detail.

It can be shown that

$$\frac{2n\hat{\theta}}{\theta} = \chi_{2n}^2$$

that is $2n\hat{\theta}/\theta$ is chi-square with $2n$ degrees of freedom. This is the exact or small sample distribution for $\hat{\theta}$.

- (b) Show that $\hat{\theta}$ is an unbiased estimator for θ .

Master's Comprehensive Examination
Part I

5. Let $x_1, x_2, \dots, x_n$ denote a random sample from a Poisson distribution with mean θ , $0 < \theta < \infty$. If the loss function is squared error and the prior distribution of θ is

$$h(\theta) = \frac{\theta^{\alpha-1} e^{-\theta/\beta}}{\Gamma(\alpha)\beta^\alpha} \quad 0 < \theta < \infty$$

(a) Find the Bayes estimator for θ .

(b) Is it unbiased?

6. $f_0(x) = \frac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}$

and

$$f_1(x) = \frac{e^{-\frac{1}{2}(x-1)^2}}{\sqrt{2\pi}}$$

Let $f = f_0$ be the null hypothesis and $f = f_1$ the alternative. A critical region, $-\frac{1}{2} < x < 0$, is selected and a single observation taken.

- (a) What is α ?
 (b) What is the power of the test?
 (c) Is this the "best" that one can do for critical regions of this size?
 (d) If the answer to (c) is yes, prove your statement. If it is no, find the "best" region and state in what sense it is best.

Mathematics

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$$\frac{1}{x^2} = x^{-2}$$

$$\frac{d}{dx} x^{-2} = -2x^{-3}$$

$$= -\frac{2}{x^3}$$

Mathematics

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Mathematics

$$\frac{1}{x^2} = x^{-2}$$

$$\frac{d}{dx} x^{-2} = -2x^{-3}$$

$$= -\frac{2}{x^3}$$

$$\frac{d}{dx} \frac{1}{x^2} = -\frac{2}{x^3}$$

$$\frac{d}{dx} \frac{1}{x^2} = -\frac{2}{x^3}$$

10/11/17

Exercise 10.10: Hypothesis Test: Standard Dev

Describe the Hypothesis
Test.

Let X be a single observation on a population with density

$$f_n(x) = \begin{cases} \frac{1}{n} & 0 \leq x \leq n \\ 0 & \text{otherwise} \end{cases}$$

Find a 95% confidence interval for θ .

UNIVERSITY OF CALIFORNIA

Master's Comprehensive Examination

Statistical Methods

December 15, 1973

Part II

Instructions: Open Book and Open Notes. Do four of the following six problems. Use a separate blue book for this section of the exam. Be sure to indicate which questions you wish graded.

1. "Our" drug company has developed a new cold relief tablet. We claim that our tablet is more effective than Brand Y in terms of the duration of the relief from the cold symptoms. To prove our claim an experiment is conducted. Twenty people are selected who have colds. Ten (10) are given "Our" drug and ten (10) are given the Brand Y drug. The response that is measured is the number of hours until the cold symptoms return.

$$\begin{array}{l} \text{Our Drug} \\ \bar{X}_O = 10 \end{array}$$

$$\Sigma X_O = 100$$

$$\frac{(\Sigma X_O)^2}{10} = 1000$$

$$\Sigma X_O^2 = 1054$$

$$n = 10$$

Brand Y Drug

$$\bar{X}_Y = 8$$

$$\Sigma X_Y = 80$$

$$\frac{(\Sigma X_Y)^2}{10} = 640$$

$$\Sigma X_Y^2 = 576$$

$$n = 10$$

- a. Does the data justify our claim? Explain how you have drawn your conclusion.
 - b. To demonstrate that "Our" drug is better than the Brand Y drug with $\alpha = .01$ what sample size would you recommend?
2. A friend claims to have N.S.P. (extra-sensory perception). He states that he can predict the result of a toss of a coin with more frequent success than chance would allow.
- We believe he is only guessing and decide to test him with 15 flips of a fair coin. Before each flip he "predicts" the result, then we flip the coin and record whether or not he was correct. Assume the trials are independent and identical.
- The following questions may be answered using normal approximation.
- a. How many correct predictions must he make in 15 trials to persuade us ($\alpha = .06$) that he has these remarkable powers?

H_0 : He is guessing

H_a : He has N.S.P.

- b. If in fact our friend actually can correctly predict the outcome with probability .80, what is the probability that the test you just constructed in part (a) will result in a Type II error?
- c. Suppose our friend makes 35 correct predictions in 50 trials. Obtain a 95% confidence interval for the true probability that he correctly predicts the outcome of a flip of a coin.

3. In order to characterize the effect of food consumption (X) on the gain in weight (Y) of animals on a certain diet twenty (20) rats were individually caged and their grams of feed consumed and grams of weight gains were measured for the interval of interest. The following calculations have been made:

$$\Sigma Y = 1200$$

$$\Sigma Y^2 = 600$$

$$\Sigma X = 400$$

$$\Sigma XY = 180$$

$$\Sigma X^2 = 90$$

- a. What is the correlation between feed consumption and weight gain?
- b. Complete the regression analysis, testing the significance of the regression of gain on feed consumption.
- c. Compute the regression equation for the prediction of weight gain from food consumption.
- d. What are the 95% probability limits for the mean weight gain when the food consumption is at the mean for this study?
- e. Construct a 95% confidence interval for the regression coefficient.

4. An experimenter in a shipyard was interested in testing the effectiveness of 5 barnacle resistant paints. Four ships were chosen for the experiment. An experimental area on the hull of each ship was defined. The area was then divided into 5 sections and the 5 paints were then randomly assigned to a section. After 6 months in the water, the barnacles were scraped from 3 sample areas (1 square foot) within each of the 5 sections.

- a. What is the model for this experiment?
- b. Write out the ANOVA table including the E.M.S.
- c. What is the appropriate test for differences among paints?

5. A manufacturer has been producing a product for several years. He has established that, with respect to a certain characteristic $\mu = 10$ and $\sigma^2 = 16$. Recently, however, the manufacturer has reason to believe that the mean of this characteristic has changed. To examine this problem a sample of size nine (9) is taken from the production line. The following was found:

$$\Sigma X = 72 \quad \bar{X} = 8 \quad n = 9$$

$$\frac{(\Sigma X)^2}{9} = 756 \quad \Sigma X^2 = 648$$

- Obtain a 95% confidence interval for μ .
 - Determine if there is sufficient evidence to demonstrate that μ has changed. Let $\alpha = .05$.
 - If the manufacturer claimed that the mean level of the process had changed, based on the above sample what would his probability of error be?
6. A family with 3 children could have 0,1,2, or 3 boys. Suppose a random sample of 40 families yielded the following data

<u>Number of Boys</u>	<u>Families Having This Number of Boys</u>
0	6
1	20
2	10
3	4

Conduct a test of hypothesis to determine whether or not it is reasonable to believe that these data represent the results of repeated sampling (40 observations) from a binomial distribution with $n=3$ and $p=\frac{1}{2}$.

RUTGERS-THE STATE UNIVERSITY
Master's Comprehensive Examination

March 17, 1973

Part I

Theory

Instructions: Open Book and Open Notes. Do four of the following six problems. Use a separate blue book for this section of the exam. Be sure to indicate which questions you wish graded.

-
1. Let the joint density of (X,Y) be given by $f(x,y) = \begin{cases} 2 & 0 \leq y \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$

Find the density of $\max(2Y,X)$.

2. Let X_1, X_2 be independent identically distributed random variables with normal distributions with variances equal to 1. Find the most powerful test of $H_0: \mu = -1$ vs. $H_1: \mu = 0$ at level $\alpha = .05$ (μ =mean of the distribution).

3. Let $X_1, X_2, \dots, X_n$ be independent identically distributed random variables such that $P[X_1 = 1] = p = 1 - P[X = 0]$. Find the maximum likelihood estimator of p^2 . Is the estimator unbiased?

4. Let $X_1, \dots, X_n$ be independent identically distributed random variables with a poisson distribution with unknown parameter λ . Find the Bayes estimator of λ with respect to the prior distribution which has density

$$g(\lambda) = \frac{1}{3!} \lambda^3 e^{-\lambda}$$

5. Let X be normal with unknown mean μ and variance 1 and Z be normal with mean $\mu+1$ and variance 2. Find that estimator of μ which minimizes the variance among unbiased linear estimators.

6. One box contains 2 dimes and a quarter and another box contains 1 dime and 1 quarter. One of the boxes is chosen at random and a coin is chosen at random from the chosen box. A second coin is then chosen from the second box. If the first coin is a dime, what is the probability that the second coin is a dime?

UNIVERSITY OF THE STATE UNIVERSITY

Master's Comprehensive Examination

March 17, 1973

Part II

Statistical Methods

Instructions: Open Book and Open Notes. Do four of the following six problems. Use a separate blue book for this section of the exam. Be sure to indicate which questions you wish graded.

1. A biologist is seeking a way of distinguishing between two organisms A and B. He decides to seek a growing medium in which the two organisms would have sharply different growing rates. He runs the following experiment.

	Nutrient					
	1		2		3	
	Organism		Organism		Organism	
	A	B	A	B	A	B
Day 1	3	4	1	6	7	6
2	6	9	2	8	10	8

where the response y_i is the volume of growth

Nutrient and organism may be treated as fixed variables. He is only able to prepare 6 agar dishes per day, so he decides to include day 1 and 2 as a block effect.

Some calculations

$$\sum y_i^2 = 440$$

$$\sum y_i = 66$$

$$(\sum y_i)^2 / n = 363$$

$$\begin{aligned} \text{TN}_1 &= 13 & \text{TO}_A &= 29 & \text{TB}_1 &= 27 & 2^2 + 9^2 + 3^2 + 11^2 + 27^2 + 11^2 &= 952 \\ \text{TN}_2 &= 17 & \text{TO}_B &= 37 & \text{TB}_2 &= 39 \\ \text{TN}_3 &= 31 \end{aligned}$$

$$\frac{(13)^2 + (17)^2 + (31)^2}{4} = 393.5$$

$$\frac{(29)^2 + (37)^2}{6} = 375$$

$$\frac{(29)^2 + (37)^2}{6} = 368.3$$

- (a) Construct the ANOVA table including the E.M.S. column. Show calculations.
- (b) Relate the question asked by the biologist to the ANOVA table and to the original data.
- (c) Does he have a method?
- (d) Calculate a confidence interval for organism A.

2. In a triangular test for selecting judges to compose a taste, a prospective judge was successful in selecting the odd sample 11 times in 15 trials. Would you select him for the panel? How many would he have to pick correctly to be chosen? What is the probability of a Type I error if we accept the above judge for our panel?
3. It is believed that the relationship between the % of catalyst in a chemical process and the yield is linear. From the data given below
 - (a) Fit the least square regression line.
 - (b) Test for linearity.
 - (c) Describe how one would proceed if the linear model is inadequate.

<u>% Catalyst = x</u>	<u>Yield = y</u>
4	70
5	71
6	73
7	74
7	73
8	76
8	75
9	77
10	72
11	81

$$\begin{aligned} \sum x &= 75 & \sum y &= 749 \\ \sum xy &= 5684 & \sum y^2 &= 55207 \\ \sum x^2 &= 605 \end{aligned}$$

4. From some development data, it has been estimated that the Power Output one one cubic component has an average, $\bar{P}_1 = 1500$ and $\sigma_1 = 30$; and a second component has $\bar{P}_2 = 2000$ and $\sigma_2 = 40$. The Power Output, $\bar{P}_3$, of the two components is additive in the net.
 - (a) What predictions can you make about rejects in production if they produce components with the above averages and standard deviations and the upper specification on $\bar{P}_3$ is 3575 and the lower specification is 3400?

- (b) What confidence limits would you place on $\bar{P}_0$ if the initial averages and standard deviations were estimated from samples of size $n = 20$?
- (c) Would any other assumptions be helpful in supporting your predictions? (If "yes", please discuss.)

5. In order to determine the effect of dietary iron sources on the hemoglobin content of the blood of mice, four experimental diets were fed to weanling mice, with ten (10) mice assigned at random to each diet. After an appropriate time on the diets a sample of blood was taken from each mouse and the forty (40) samples were subjected to duplicate analyses for hemoglobin.

- (a) Make a schematic representation of this experimental plan.
- (b) Complete the following analysis of variance and write out the expected values of the mean squares:

Source of Variation	d.f.	Sum of Squares	M.S.	E.M.S.
Total	79	6360		
Diets		1800		
Mice within Diets			100	
Analyses within Mice within Diets				

- (c) Test the significance of the effect of the diets on the hemoglobin content of the blood and make a probability statement of interpretation.
- (d) Which is the greater source of experimental variation -- Mice or Analyses? Is the difference in variability due to these sources statistically significant? Support your conclusion with a probability statement.
- (e) What magnitude of difference in the hemoglobin content of two of the dietary means would be needed to be considered "significant" at the 1% probability level?

6. A quality control engineer desires to characterize the variability in a process which is carried out by a large number of automatic machines running continuously. He suspects that the greatest variation is from machines, but is also reasonably certain that the production from any one machine may not be statistically uniform. He knows nothing of the variability of the test method but suspects it will be relatively small compared to product variation and is not willing to spend much effort to characterize it. He has two weeks in which to complete this study and can spend money for sixty (60) tests.

- (a) Propose a design which will best satisfy his needs. Show schematically.
- (b) Write out the analysis of variance for your design with the expected values of the mean squares using general notation if the design you propose is unbalanced.

RUTGERS-THE STATE UNIVERSITY
Master's Comprehensive Examination

December 16, 1972

Theory

Part I

Instructions: Open Book and Open Notes. Do four of the following six problems. Use a separate blue book for this section of the exam. Be sure to indicate which questions you wish graded.

1. Let $X_1, \dots, X_n$ be independent with X_i having a normal distribution with mean $\mu_i = \theta + C_i$, $i=1, \dots, n$, where C_i are known constants and θ and τ are unknown.
 - (a) Find a sufficient statistic for the pair (θ, τ) .
 - (b) Find the maximum likelihood estimators of θ and τ .
 - (c) What is the distribution of the maximum likelihood estimator of θ ?
2. Let $X_1, \dots, X_n$ be a random sample of size n from a population with density function

$$f_\theta(x) = \frac{x^n}{n! \theta^{n+1}} e^{-x/\theta}, \quad x > 0, \quad \theta > 0$$

- (a) Find a sufficient statistic for θ .
 - (b) Is there a uniformly minimum variance unbiased estimator for θ ? If so, find it.
 - (c) Find the maximum likelihood estimator of θ .
3. Let $X_1, \dots, X_n$ be independent identically distributed random variables such that

$$P_\lambda \{X_1 = j\} = \frac{e^{-\lambda} \lambda^j}{j!}, \quad j = 0, 1, \dots, \lambda > 0$$

It is desired to test the hypothesis $H_0: \lambda = 1$ vs. $H_1: \lambda = \frac{1}{2}$. Find the most powerful test of H_0 vs. H_1 at level of significance $\alpha = e^{-n}$.

Part I

4. Let the continuous random variable X have a uniform distribution on $[0,1]$ and let c be an constant such that $0 \leq c \leq 1$. Define the following random variables:

$$Y_1 = \begin{cases} 1 & \text{if } X \leq c \\ 0 & \text{otherwise} \end{cases}$$

$$Z = \begin{cases} \sqrt{X} & \text{if } X \leq c \\ X - c(1-c) & \text{if } X > c \end{cases}$$

$$Y_2 = \begin{cases} 1 & \text{if } Z < c \\ 0 & \text{otherwise} \end{cases}$$

Show that Y_1 and Y_2 are independent and identically distributed.

5. Ten pairs of gloves were randomly placed in a washing machine to be cleaned. Unfortunately, on removal from the machine, 3 gloves were found to be stained. A pair of gloves is considered to be ruined if either one or both gloves are stained. Let X = number of ruined pairs of gloves. Find $E[X]$.

6. Let the continuous random variables X and Y have joint density function

$$f(x,y) = \begin{cases} 2 & \text{if } 0 \leq x \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Let $Z = \max(2X, Y)$. Find the density function for the random variable Z .

RUTGERS--THE STATE UNIVERSITY
Master's Comprehensive Examination

December 16, 1972

Part II

Statistical Methods

Instructions: Open Book and Open Notes. Do four of the following six problems. Use a separate blue book for this section of the exam. Be sure to indicate which questions you wish graded.

1. A four source balanced nested design was conducted. (The four sources will be labeled A, B, C, D). The resulting ANOVA table was developed.

<u>Source</u>	<u>df</u>	<u>SS</u>	<u>ms</u>	<u>ms</u>
Total	15			
A	1		8.00	
B(A)	2		4.00	
C(B(A))	4		2.00	
D(C(B(A)))	8		0.50	

- (a) Make a schematic of the experimental design.
 - (b) Complete the above ANOVA table.
 - (c) Indicate what F tests may be performed and hypothesis that they would test.
 - (d) Estimate the variability of the following
 - (i) The individual observations about the mean of B.
 - (ii) The mean of C about the grand average where the mean of C is found from two levels of D.
 - (iii) The grand average.
2. Six (6) individuals were selected to take a typing test. The test was conducted by giving each person one page of material to type. The data collected was the amount of time needed to type the page as well as the number of words incorrectly typed. The results are shown below.

<u>Individual</u>	<u>No. of Errors</u>	<u>Amount of Time (Sec.)</u>
1	2	62
2	3	61
3	1	65
4	20	120
5	6	55
6	4	57

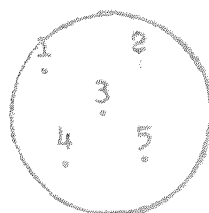
Completely analyze the data.

Part II

3. It is desired to determine if a certain drug has any effect on the red cell count in the blood of men. Five men are selected at random and their red cell count is observed. These men are then given the drug. Five days later their red cell count is observed again. The data is shown below.

Individual	Before Drug is Administered	5 Days After Drug is Administered
1	4	5
2	2	1
3	1	2
4	3	4
5	7	8

- (a) Perform an ANOVA to determine if there exists any difference between the two population means ($\alpha = .05$)
- (b) Use a t-test to compare the two means ($\alpha = .05$).
- (c) Do the methods in (a) and (b) agree? If so, why; if not, why not?
- (d) Which of the two methods is better? Explain.
4. It is desired to determine the effect of both depth and position in a beaker on the concentration of a certain solution. The concentration is measured at three depths from the surface of the beaker. At each depth, measurements are taken at five specific lateral positions in the beaker. The positions are shown below.



At each depth and position combination two observations are made.

- (a) Write out the analysis of variance table including the sources of variability, the degrees of freedom, and the expected value of the mean squares.
- (b) Suppose the above experiment was changed so that the five lateral positions at each depth were selected entirely at random. Repeat part (a).
- (c) How will the interpretations resulting from the experiments shown in (a) and (b) compare?

Part II

5. Three hundred (300) individuals were polled about their brand of toothpaste at two different times, (now and a year ago). The poll results are summarized in the following table.

	<u>Now</u>	
	A	B
<u>Last Year</u>	A 180	20
	B 10	90

Test, at $\alpha = .05$, the null hypothesis that an individual who switched brands over that time period is as likely to have switched to brand A as to brand B.

6. It was desired to conduct an experiment to compare the effects of length of cold storage on the tenderness of beef roasts. There were six (6) periods of storage to be denoted $T_1, \dots, T_6$. Thirty (30) roasts from the round of an animal were available. The roasts on an animal group themselves naturally into pairs, since to each roast on the left side of an animal there corresponds another roast on the right side. From previous experience it was believed that the two roasts in any pair would give closely similar results. Also the variation between the pair of roasts would be considerably larger. Only one animal was available for the experiment.

Each roast was to be evaluated by four judges on a scale from 0 to 10. The responses used would then be the total of the scores from the four judges.

- Give the name of the experimental design that you would recommend to the experimenter. Indicate why you feel the design is the best to use.
- Indicate how the experimental treatments should be allocated and be specific.
- Give the ANOVA table including the sources of variability, the degrees of freedom, and the e.m.s.
- Indicate the hypotheses of interest to the experimenter and how these hypotheses can be tested.
- Briefly indicate what other procedures you might use to help interpret the results of the experiment.

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

Part I

Theory

March 18, 1972

Instructions: Open Book and Open Notes. Do four of the following six problems. Use a separate blue book for this section of the exam. Be sure to indicate which questions you wish graded.

1. A test is proposed which is claimed to be 95% effective in detecting the presence of a certain disease; that is, if D is the event that a person has the disease and P is the event that a person's test is positive, $P(P|D) = .95$ and $P(\text{Not } D|\text{Not } P) = .95$.

If the incidence of this disease in the population is such that $P(D) = 0.001$ and a mass testing program is carried out, on the average what percent of "positives" will actually have the disease?

2. A urn contains chips numbered $1, 2, \dots, N$. Draw n chips without replacement ($n < N$).

- (a) What is the probability that the smallest number drawn is j ?
(b) What is the probability that the second smallest is j ?

3.
$$f_0(x) = \frac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}$$

and

$$f_1(x) = \frac{e^{-\frac{1}{2}(x-1)^2}}{\sqrt{2\pi}}$$

Let $f = f_0$ be the null hypothesis and $f = f_1$ the alternative. A critical region, $-\frac{1}{2} < x < 0$, is selected and a single observation taken.

- (a) What is α ?
(b) What is β ?
(c) What is the power of the test?
(d) Is this the "best" that one can do for critical regions of this size?
(e) If the answer to (d) is yes, prove your statement. If it is no, find the "best" region and state in what sense it is best.

[illegible]

Journal of Management Studies, 19(6), 701-718.

1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30. 31. 32. 33. 34. 35. 36. 37. 38. 39. 40. 41. 42. 43. 44. 45. 46. 47. 48. 49. 50. 51. 52. 53. 54. 55. 56. 57. 58. 59. 60. 61. 62. 63. 64. 65. 66. 67. 68. 69. 70. 71. 72. 73. 74. 75. 76. 77. 78. 79. 80. 81. 82. 83. 84. 85. 86. 87. 88. 89. 90. 91. 92. 93. 94. 95. 96. 97. 98. 99. 100. 101. 102. 103. 104. 105. 106. 107. 108. 109. 110. 111. 112. 113. 114. 115. 116. 117. 118. 119. 120. 121. 122. 123. 124. 125. 126. 127. 128. 129. 130. 131. 132. 133. 134. 135. 136. 137. 138. 139. 140. 141. 142. 143. 144. 145. 146. 147. 148. 149. 150. 151. 152. 153. 154. 155. 156. 157. 158. 159. 160. 161. 162. 163. 164. 165. 166. 167. 168. 169. 170. 171. 172. 173. 174. 175. 176. 177. 178. 179. 180. 181. 182. 183. 184. 185. 186. 187. 188. 189. 190. 191. 192. 193. 194. 195. 196. 197. 198. 199. 200. 201. 202. 203. 204. 205. 206. 207. 208. 209. 210. 211. 212. 213. 214. 215. 216. 217. 218. 219. 220. 221. 222. 223. 224. 225. 226. 227. 228. 229. 230. 231. 232. 233. 234. 235. 236. 237. 238. 239. 240. 241. 242. 243. 244. 245. 246. 247. 248. 249. 250. 251. 252. 253. 254. 255. 256. 257. 258. 259. 260. 261. 262. 263. 264. 265. 266. 267. 268. 269. 270. 271. 272. 273. 274. 275. 276. 277. 278. 279. 280. 281. 282. 283. 284. 285. 286. 287. 288. 289. 290. 291. 292. 293. 294. 295. 296. 297. 298. 299. 300. 301. 302. 303. 304. 305. 306. 307. 308. 309. 310. 311. 312. 313. 314. 315. 316. 317. 318. 319. 320. 321. 322. 323. 324. 325. 326. 327. 328. 329. 330. 331. 332. 333. 334. 335. 336. 337. 338. 339. 340. 341. 342. 343. 344. 345. 346. 347. 348. 349. 350. 351. 352. 353. 354. 355. 356. 357. 358. 359. 360. 361. 362. 363. 364. 365. 366. 367. 368. 369. 370. 371. 372. 373. 374. 375. 376. 377. 378. 379. 380. 381. 382. 383. 384. 385. 386. 387. 388. 389. 390. 391. 392. 393. 394. 395. 396. 397. 398. 399. 400. 401. 402. 403. 404. 405. 406. 407. 408. 409. 410. 411. 412. 413. 414. 415. 416. 417. 418. 419. 420. 421. 422. 423. 424. 425. 426. 427. 428. 429. 430. 431. 432. 433. 434. 435. 436. 437. 438. 439. 440. 441. 442. 443. 444. 445. 446. 447. 448. 449. 450. 451. 452. 453. 454. 455. 456. 457. 458. 459. 460. 461. 462. 463. 464. 465. 466. 467. 468. 469. 470. 471. 472. 473. 474. 475. 476. 477. 478. 479. 480. 481. 482. 483. 484. 485. 486. 487. 488. 489. 490. 491. 492. 493. 494. 495. 496. 497. 498. 499. 500. 501. 502. 503. 504. 505. 506. 507. 508. 509. 510. 511. 512. 513. 514. 515. 516. 517. 518. 519. 520. 521. 522. 523. 524. 525. 526. 527. 528. 529. 530. 531. 532. 533. 534. 535. 536. 537. 538. 539. 540. 541. 542. 543. 544. 545. 546. 547. 548. 549. 550. 551. 552. 553. 554. 555. 556. 557. 558. 559. 560. 561. 562. 563. 564. 565. 566. 567. 568. 569. 570. 571. 572. 573. 574. 575. 576. 577. 578. 579. 580. 581. 582. 583. 584. 585. 586. 587. 588. 589. 590. 591. 592. 593. 594. 595. 596. 597. 598. 599. 600. 601. 602. 603. 604. 605. 606. 607. 608. 609. 610. 611. 612. 613. 614. 615. 616. 617. 618. 619. 620. 621. 622. 623. 624. 625. 626. 627. 628. 629. 630. 631. 632. 633. 634. 635. 636. 637. 638. 639. 640. 641. 642. 643. 644. 645. 646. 647. 648. 649. 650. 651. 652. 653. 654. 655. 656. 657. 658. 659. 660. 661. 662. 663. 664. 665. 666. 667. 668. 669. 670. 671. 672. 673. 674. 675. 676. 677. 678. 679. 680. 681. 682. 683. 684. 685. 686. 687. 688. 689. 690. 691. 692. 693. 694. 695. 696. 697. 698. 699. 700. 701. 702. 703. 704. 705. 706. 707. 708. 709. 710. 711. 712. 713. 714. 715. 716. 717. 718. 719. 720. 721. 722. 723. 724. 725. 726. 727. 728. 729. 730. 731. 732. 733. 734. 735. 736. 737. 738. 739. 740. 741. 742. 743. 744. 745. 746. 747. 748. 749. 750. 751. 752. 753. 754. 755. 756. 757. 758. 759. 760. 761. 762. 763. 764. 765. 766. 767. 768. 769. 770. 771. 772. 773. 774. 775. 776. 777. 778. 779. 780. 781. 782. 783. 784. 785. 786. 787. 788. 789. 790. 791. 792. 793. 794. 795. 796. 797. 798. 799. 800. 801. 802. 803. 804. 805. 806. 807. 808. 809. 810. 811. 812. 813. 814. 815. 816. 817. 818. 819. 820. 821. 822. 823. 824. 825. 826. 827. 828. 829. 830. 831. 832. 833. 834. 835. 836. 837. 838. 839. 840.

Figure 1. The effect of the concentration of the *Agrobacterium* suspension on the transformation efficiency of *Agrobacterium* strains.

1. *Chlorophyll a* and *Chlorophyll b* were determined by the method of Arar and Collins (1987).

1. 2. 3.

4. Let $x_1, x_2 \dots x_n$ denote a random sample from a Poisson distribution with mean θ , $0 < \theta < \infty$. If the loss function is squared error and the prior distribution of θ is

$$h(\theta) = \frac{\theta^{\alpha-1} e^{-\theta/\beta}}{\Gamma(\alpha)\beta^\alpha} \quad 0 < \theta < \infty$$

- (a) Find the Bayes estimator for θ .
- (b) Is it consistent?
- (c) Is it unbiased?

5. Independent samples of sizes n_1 and n_2 , respectively, are drawn from two normal populations with equal means μ and with variances equal to σ^2 and $\lambda\sigma^2$.

- (a) Find the maximum likelihood estimate of μ if we use the information in both samples.
- (b) Show that this estimate has variance

$$\frac{\sigma^2}{(n_1 + \frac{n_2}{\lambda})}$$

6. X_1, X_2 are two independent observations from a uniform distribution in the interval $(0, \theta)$. Show that $(Y, \frac{Y}{\sqrt{\alpha}})$ is a confidence interval for θ with confidence coefficient $1 - \alpha$, where $Y = \max(X_1, X_2)$.

UNIVERSITY OF MICHIGAN LIBRARY

Michigan's Comprehensive Examination

Statistical Methods

March 18, 1972

Part II

Instructions: Open Book and Open Notes. Do four of the following six problems. Use a separate blue book for this section of the exam. Be sure to indicate which questions you wish graded.

1. From the simple linear regression model

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

the following results were obtained

$$\begin{aligned} n &= 11 & \sum (X_i - \bar{X})^2 &= 64 \\ \bar{X} &= 4 & \sum (X_i - \bar{X})(Y_i - \bar{Y}) &= 256 \\ \bar{Y} &= 20 & \sum (Y_i - \bar{Y})^2 &= 1600 \end{aligned}$$

- (a) Find the estimated regression line.
 - (b) Find the minimum variance unbiased estimate of σ^2 .
 - (c) Test the hypothesis that $\beta_1 = 0$.
Let the probability of a Type I error = .10.
 - (d) Find a 95% confidence interval for a value of Y observed at some time in the future for $X = 4$.
 - (e) Find an estimate of the correlation coefficient ρ .
 - (f) Test the hypothesis that $\rho = .5$.
2. From some development data, it has been estimated that the Power Output on one radio component has an average, $\bar{P}_1 = 1500$ and $\sigma_1 = 30$; and a second component has $\bar{P}_2 = 2000$ and $\sigma_2 = 40$. The Power Output, $\bar{P}_3$, of the two components is additive in the test.
- (a) What prediction can you make about rejects in production if they produce components with the above averages and standard deviations and the upper specification on $\bar{P}_3$ is 3575 and the lower specification is 3400?
 - (b) What confidence limits would you place on $\bar{P}_3$ if the initial averages and standard deviations were estimated from samples of size $n = 25$?
 - (c) Would any other assumptions be helpful in supporting your predictions? (If "yes", please discuss.)

3. Upon calculating the analysis of variance of the pumping rates of oysters in six different water temperatures the corrected sum of squares for each of the four sources of variation (appropriate to the design) was found to be 245. The design was a randomized block in which eight different oysters were taken, two being assigned to each of the water temperature baths.

- (a) Make an abbreviated schematic of the design.
- (b) Complete the following analysis of variance showing the expected values of the mean squares.

| Source of Variation | D.F. | Sum of Squares | M.S. | E.M.S. |
|---------------------|------|----------------|------|--------|
| Total | | | | |
| Temperature | | | | |
| Beds | | | | |
| Temp. x Beds | | | | |
| Oysters within Bed | | | | |

- (c) Test the significance of the temperature effect and make a probability statement of interpretation.
- (d) The two highest temperatures gave pumping rates of 41.7 and 44.7 (gallons per hour) -- the later being for the highest temperature. At this range of temperatures has increasing the temperature significantly increased the pumping rate of the oyster? Support your answer with a probability statement.

4. A quality control engineer desires to characterize the variability in a process which is carried out by a large number of automatic machines running continuously. He suspects that the greatest variation is among machines but is also reasonably certain that the production from any one machine may not be consistently uniform. He knows nothing of the variability of the testing method but suspects it will be relatively small compared to product variation and is not willing to spend much effort to characterize it. He has two weeks in which to complete this study and can expend money from sixty (60) tests.

- (a) Propose a design which will best satisfy his needs. Show schematically.
- (b) Write out the analysis of variance for your design with the expected values of the mean squares using general notation if the design you propose is unbalanced.

5. Measurements of the yields of two processes, A and B, have shown average yields of $\bar{X}_A = 46.5$, $\bar{X}_B = 47.1$. The process yields are independent normal

variables with means μ_A and μ_B and variances, $\sigma_A^2 = 25$ and $\sigma_B^2 = 16$ respectively. $\bar{X}_A$ is based on 100 observations and $\bar{X}_B$ on 64. We want to test $H_0: \mu_A = \mu_B$ with a 2-sided, 10% level test.

- (a) Give the Critical Region of the test.
- (b) Carry out the test for the data above.
- (c) If $\mu_A = 50$ and $\mu_B = 48$, what is the power of this test?
- (d) Give a 90% confidence interval for $(\mu_A - \mu_B)$.

6. Assume that 2 factors A and B have been included in an experiment, each factor at 2 levels. Construct a two way table (and a graph) using the numbers (24, 37, 57, 70) to illustrate each of the following hypothetical situations. (Note: you may use a number more than once in the same table).

- (a) No A effect, Significant B effect, Zero interaction.
- (b) Significant A effect, Significant B effect, Zero interaction.
- (c) No A or B effects, Significant interaction.

Rutgers-The State University
Master's Comprehensive Examination

Part I

Theory

December 18, 1971

9:00 - 12:00

Instructions: (a) Open Book, Open Notes
(b) Use a separate blue book for each Part.
(c) Answer 4 out of 6 questions. Indicate on the front of your blue book which questions you wish graded.

1. Let $X_1, X_2, \dots$ be independent random variables with $P[X_1 = 1] = \frac{1}{2} = P[X_1 = 0]$ for all i .

(a) Find

$$\lim_{n \rightarrow \infty} P[(X_1 + \dots + X_n)/n \leq 1]$$

(b) Find

$$\lim_{n \rightarrow \infty} P[(X_1 + \dots + X_n)^2/n \leq 1]$$

(c) Find

$$\lim_{n \rightarrow \infty} P\left[\frac{X_1 - X_2 + X_3 - X_4 + \dots - X_{2n}}{\sqrt{n}} \leq 1\right]$$

(d) Find

$$\lim_{n \rightarrow \infty} P\left[\frac{X_1 + \dots + X_n}{n^{\frac{1}{4}}} \leq 1\right]$$

2. A fish breeder is breeding fancy guppies. He knows that a mother guppy can give birth to large numbers of babies at a time, but not all of these baby guppies survive. He is interested in modeling the distribution of the number of guppies that actually survive.

Suppose that the number of guppies born is a Poisson variate with mean 30. Suppose, in addition, that each guppy born (independently of others) has a one-half chance of survival.

- (a) Using conditional expectations, find the expected number of guppies surviving, and the variance of the number of guppies surviving.
- (b) Find the probability that exactly r guppies survive.
- (c) Calculate (approximate) the probability that at least 20 guppies survive. Describe your approximation, and the basis for using it.

Part I

3. We know that a coin tossing experiment was conducted, but, we only have partial results. We know that the coin was fair and that 3 tails were observed.
- What is a M.L.E. of the total number of tosses (show)?
 - Is the M.L.E. unique?
 - Find an unbiased estimator for the general case where we know that t tails were observed.
4. A coin that has probability θ of coming up heads is tossed repeatedly until a head actually occurs. Let T = number of tails observed in this procedure.
- Find a sufficient statistic for θ .
 - Find the M.L.E. of θ .
 - Is there a U.M.P. test of level $1/16$ for $H_0: \theta = \frac{1}{2}$ vs. $H_1: \theta > \frac{1}{2}$? If so, find it.
5. Suppose that $X_1, \dots, X_n$ are independently and identically distributed according to the uniform distribution on $[0, \theta]$.
- Let $M_n = \max(X_1, \dots, X_n)$
- Let a test of $H_0: \theta \leq \theta_0$ vs. $H_1: \theta > \theta_0$ be given by
- Reject if $M_n(X_1, \dots, X_n) > C$.
- Compute the power function of the test and show that it is monotone increasing in θ_0 .
 - In testing $H_0: \theta \leq \frac{1}{2}$ vs. $H_1: \theta > \frac{1}{2}$ what choice of C would make the size exactly .05?
 - How large should n be so that the power of the test in (b) has power at least .98 for $\theta = \frac{3}{4}$?

Part I

6. If $X_1, \dots, X_4$ are independently distributed with density

$$P(x) = \begin{cases} \frac{1}{\Gamma(3)\beta^3} x^2 e^{-x/\beta} & x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

(a) Find the MLE of β .

(b) What is the distribution of $2 \frac{\hat{\beta}}{\beta}$?

(c) Using your answer in (b) construct a 90% confidence interval for β if the data are

$$X_1 = 3 \quad X_2 = 4 \quad X_3 = 5 \quad X_4 = 1.$$

ROCHESTER STATE UNIVERSITY

Master's Comprehensive Examination

Statistical Methods

December 18, 1971

Part II

Instructions: Open Book and Open Notes. Do four of the following six problems. Use a separate blue book for this section of the exam. Be sure to indicate which questions you wish graded.

1. A new packaging material was compared against a standard material. The test was conducted by storing the same food product in each of these for a fixed period of time and then determining the moisture loss. Each material was used to wrap ten (10) food samples. The following results were obtained.

| | <u>New Material</u> | <u>Old Material</u> |
|--------------------------|---------------------|---------------------|
| Sample size | 10 | 10 |
| Mean moisture loss | 224 | 232 |
| $\sum (x_i - \bar{x})^2$ | 300 | 400 |

- (a) Conduct a statistical test to determine if the new material had a lower mean moisture loss than the old material. Let $\alpha = .05$.
- (b) Find a 95% confidence interval for the difference between the two mean moisture losses.
2. (a) Compute Y's regression on X using the following data and the model $E(Y) = \beta_0 + \beta_1 X$.

| | | | | | | | |
|---|----|----|----|---|---|---|---|
| Y | 2 | 3 | 4 | 4 | 5 | 7 | 7 |
| X | -3 | -2 | -1 | 0 | 1 | 2 | 3 |

- (b) Is the slope significantly different from zero?
- (c) Find the 95 percent confidence interval for the $E(Y)$ when $X=1$.
3. In order to determine the effect of dietary iron sources on the hemoglobin content of the blood of mice, four experimental diets were fed to weanling mice, with ten (10) mice assigned at random to each diet. After an appropriate time on the diets a sample of blood was taken from each mouse and the forty (40) samples were subjected to duplicate analyses for hemoglobin.
- (a) Make a schematic representation of this experimental plan.

Part XI

- (b) Complete the following analysis of variance and write out the expected values of the mean squares:

| <u>Source of Variation</u> | <u>d.f.</u> | <u>Sum of Squares</u> | <u>M.S.</u> | <u>E.M.S.</u> |
|--------------------------------------|-------------|-----------------------|-------------|---------------|
| Total | 79 | 6360 | | |
| Diets | | 1800 | | |
| Mice within Diets | | | 100 | |
| Analyses within Mice
within Diets | | | | |

- (c) Test the significance of the effect of the diets on the hemoglobin content of the blood and make a probability statement of interpretation.
- (d) Which is the greater source of experimental variation — Mice or Analyses? Is the difference in variability due to these sources statistically significant? Support your conclusion with a probability statement.
- (e) What magnitude of difference in the hemoglobin content of two of the dietary means would be needed to be considered "significant at the 1% probability level"?
- (f) What is the standard deviation due to analysis?

Part II

4. A small bottle (sample) from each of 25 consecutive production batches was analyzed by each of two analysts, A and B.

| Present evidence pertaining to such important matters as: | <u>Batch No.</u> <u>Analyst A</u> <u>Analyst B</u> | | |
|---|--|----|----|
| | | | |
| (a) stability of the production process, and | 1 | 93 | 93 |
| | 2 | 89 | 91 |
| | 3 | 84 | 93 |
| | 4 | 90 | 90 |
| | 5 | 87 | 87 |
| (b) agreement between the two analysts. | 6 | 90 | 90 |
| | 7 | 93 | 92 |
| | 8 | 89 | 89 |
| | 9 | 89 | 90 |
| | 10 | 90 | 90 |
| (c) Discuss implications. | 11 | 90 | 90 |
| | 12 | 92 | 89 |
| | 13 | 92 | 88 |
| | 14 | 92 | 92 |
| | 15 | 92 | 93 |
| | 16 | 90 | 90 |
| | 17 | 90 | 90 |
| | 18 | 90 | 90 |
| | 19 | 87 | 92 |
| | 20 | 92 | 93 |
| | 21 | 88 | 88 |
| | 22 | 89 | 89 |
| | 23 | 94 | 94 |
| | 24 | 93 | 93 |
| | 25 | 91 | 92 |

5. An experiment was conducted to determine the effect of temperature on the rate of reaction of a chemical experiment. The temperatures considered were 200 degrees C., 250 degrees C., and 300 degrees C. At each temperature four (4) experiments were run and the rate of reaction was recorded. An ANOVA was performed on this data and it was found that Error $_{MS} = 9.0$. It was also found that the average rate of reaction for each temperature was as follows.

$$T_{200} = 1$$

$$T_{250} = 4$$

$$T_{300} = 7$$

- (a) Completely write out the ANOVA table.
- (b) Perform the appropriate F test, state the null and alternative hypothesis, and state the conclusions you draw from this test.
- (c) Use the l.s.d. to compare the three treatments. What conclusions do you draw from this test?

6. From the production line for the manufacture of glass bottles a sampling study was conducted to investigate the number of defective bottles. Samples of 100 bottles were chosen at random from each of three machines on each of the three shifts on five consecutive production days.

The numbers of defective bottles/100 bottles inspected were subjected to analysis of variance appropriate to the design of the study. The following is a partial summary of the ANOVA.

| <u>Source</u> | <u>df</u> | <u>Sums of Squares</u> | <u>Mean Squares</u> | <u>E.M.S.</u> |
|---------------|-----------|------------------------|---------------------|---------------|
| Total | 44 | 1179 | | |
| Machines | | | 115.50 | |
| Dates | | 259 | | |
| MxD | | | 12.25 | |
| Shifts | | 156 | | |
| MxS | | | 2.25 | |
| DxS | | 150 | | |
| MxDxS | | | | |

- (a) Complete the ANOVA shown above assuming Machines and Dates to be Random and Shifts to be Fixed effects.
- (b) Make a complete statistical interpretation of this analysis using the following means and some sort of appropriate intervals.

| | | | | | |
|-----------|------|---------|------|-------|------|
| Machine 1 | 8.2 | Shift A | 7.9 | Day 1 | 7.7 |
| 2 | 9.5 | B | 10.9 | 2 | 7.7 |
| 3 | 13.5 | C | 12.4 | 3 | 10.7 |
| | | | | 4 | 13.7 |
| | | | | 5 | 12.3 |

Part II

- (c) In an effort to reduce variability in defects what steps would you propose?
- (d) Assuming that Machine, Date and Shift variation could be eliminated to that which might be due to random "uncontrollable" variation what is your best estimate of the standard deviation of such an improved process?

Michigan State University
Master's Examination in Statistics

March 20, 1971

Part I

- (a) Answer 4 out of 6 questions.
(b) Use separate blue books for Part I and Part II.

1. There are 3 white discs and 7 black discs in an urn. If 3 discs are withdrawn by random selection without replacement, what is the probability that 2 white discs and 1 black disc will appear?
- (b) Someone rolls a pair of dice and tells you the result is odd and exceeds 6. What is the conditional probability that it is a 7 or 11?
2. Suppose n electron tubes are placed on life test. The experiment is stopped when all tubes have failed, the times of failure being $x_1, \dots, x_n$. Assume that the time-to-failure has an exponential distribution with probability density function given by

$$f(x; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}} \quad x \geq 0, \theta > 0$$
$$= 0 \quad x < 0$$

where the parameter θ is the mean of the distribution.

- (a) Derive the maximum likelihood estimator $\hat{\theta}$ of the mean life θ . Show work in detail.

It can be shown that

$$\frac{2n\hat{\theta}^2}{\theta} = \chi_{2n}^2$$

that is $2n\hat{\theta}^2/\theta$ is chi-square with $2n$ degrees of freedom. This is the exact or small sample distribution for $\hat{\theta}$.

- (b) Show that $\hat{\theta}$ is an unbiased estimator for θ .
- (c) Find $V(\hat{\theta})$, the variance of $\hat{\theta}$.

Master's Examination
Part I

3. Let $x_1, x_2, \dots, x_n$ denote a random sample from a Poisson distribution with mean θ , $0 < \theta < \infty$. If the loss function is squared error and the prior distribution of θ is

$$h(\theta) = \frac{e^{-\theta} \theta^{a-1}}{\Gamma(a) b^a} \quad 0 < \theta < \infty$$

- (a) Find the Bayes estimator for θ .
- (b) Is it consistent?
- (c) Is it unbiased?

4. If x_1 and x_2 is a random sample of two from a population with the uniform distribution over the unit interval, show that the distribution of $u = x_1 x_2$ is

$$f(u) = -\log u \quad 0 < u < 1$$

5. X_1, X_2 are two independent observations from a uniform distribution in the interval $(0, \theta)$. Show that $(Y, \frac{Y}{\sqrt{2}})$ is a confidence interval for θ with confidence coefficient $1 - \alpha$, where $Y = \max(X_1, X_2)$.

6. $f_0(z) = \frac{e^{-\frac{1}{2}z^2}}{\sqrt{2\pi}}$

and

$$f_1(z) = \frac{e^{-\frac{1}{2}(z-1)^2}}{\sqrt{2\pi}}$$

Let $H = f_0$ be the null hypothesis and $K = f_1$ the alternative. A critical region, $-\frac{1}{2} < x < 0$, is selected and a single observation taken.

- (a) What is α ?
- (b) What is β ?
- (c) What is the power of the test?
- (d) Is this the "best" that one can do for critical regions of this size?
- (e) If the answer to (d) is yes, prove your statement. If it is no, find the "best" region and state in what way it is best.

Brigham Young University
Master's Examination in Statistics

March 20, 1971

Part II

- (a) Answer 4 out of 6 questions.
(b) Use separate blue books for Part I and Part II.

1. For the following data, make an appropriate analysis of means; use the accompanying sheet of graph paper:

| | | | | | | | | | |
|----------|----|----|----|----|---|----|----|----|----|
| 3 | 1 | 6 | 4 | 4 | 0 | 7 | 6 | 8 | 6 |
| 5 | -1 | 4 | 3 | 2 | 3 | 4 | 8 | 5 | 8 |
| 0 | 1 | -1 | 5 | 3 | 0 | 5 | 7 | 6 | 9 |
| 4 | 4 | 3 | 6 | 4 | 5 | 6 | 9 | 6 | 8 |
| 1 | 0 | 2 | 5 | 3 | 2 | 4 | 7 | 8 | 7 |
| $n = 13$ | 5 | 14 | 23 | 16 | 8 | 25 | 37 | 33 | 35 |

These readings represent (coded) tests of 5 samples from each of 10 different spools of wire. What is your interpretation?

2. The two factors in a fully-crossed, replicated experimental design are: Glass electrodes (A) at 4 levels and calomel electrodes (B) at 2 levels. Two replications will be made at each of the 8 distinct experimental conditions for a total of 16 tests. The design in blank would appear as Table I.

| | | Glass Electrodes (A) | | | |
|------------------------|-----------------|----------------------|----------------|----------------|----------------|
| | | A ₁ | A ₂ | A ₃ | A ₄ |
| Calomel Electrodes (B) | B _I | | | | |
| | B _{II} | | | | |

- (a) Describe how you would specify the order in which the 16 tests are to be made so that the full randomization required by this design will be achieved, e.g., which of the 16 runs will be responsible for the first observed result under condition A-2 and B-I, etc? Suppose the experiment has been completed and the results are as shown in Table II.

Master's Examination
Part II

March 20, 1973

| Table II | | Glass Electrodes (A) | | | |
|------------------------------|----|----------------------|---|---|---|
| | | 1 | 2 | 3 | 4 |
| Calomel
Electrodes
(B) | I | 2 | 5 | 3 | 1 |
| | | 7 | 4 | 6 | 0 |
| | | 3 | 6 | 3 | 2 |
| | II | 3 | 6 | 5 | 3 |

From these observed values, the following ANOVA Summary table can be obtained as Table III.

| Table III | | | | | |
|-------------|---------|------|----|-----|----------------|
| Source | SS | d.f. | MS | AMS | F ₀ |
| Glass (A) | 31.875 | | | | |
| Calomel (B) | 1.5625 | | | | |
| AB | 1.875 | | | | |
| Residual | 8.000 | | | | |
| Total | 43.3125 | | | | |

- (b) Fill in the d.f. and MS columns in Table III.
- (c) If the experiment is a Model 1 (fixed effects) type, fill in the AMS (expected mean square) column of Table III.
- (d) If the experiment is a Model 2 (random effects) type, fill in the AMS column of Table III.
- (e) Make the appropriate F-tests either (c) and (d) are completed and state your conclusions. (Use $\alpha = .05$).
3. Measurements of the yields of two processes, A and B, have shown average yields of $\bar{X}_A = 46.5$ $\bar{X}_B = 48.1$. The process yields are independent normal variables with means μ_A and μ_B and variances $\sigma_A^2 = 25$ and $\sigma_B^2 = 16$ respectively. $\bar{X}_A$ is based on 100 observations and $\bar{X}_B$ on 64. We want to test $H_0: \mu_A = \mu_B$ with a 2-sided, 10% level test.

Master's Examination
Part II

- (a) Give the Critical Region of the test.
(b) Carry out the test for the data above.
(c) If $\mu_A = 50$ and $\mu_B = 48$, what is the power of this test?
(d) Give a 90% confidence interval for $(\mu_A - \mu_B)$.
4. To illustrate fractional replication, suppose that we have a 2^4 experiment with factors A, B, C, D, and wish to obtain a $\frac{1}{2}$ replicate by using ABD as the defining contrast.

- (a) What treatments in addition to (1) are used?
(b) How many experimental units are used?
(c) What are the aliases of all main effects and two factor interactions?
(d) Would you prefer another interaction in place of ABD in constructing a $\frac{1}{2}$ replicate? Why?

5. In order to determine the effect of dietary iron sources on the hemoglobin content of the blood of mice, four experimental diets were fed to weanling mice, with ten (10) mice assigned at random to each diet. After an appropriate time on the diets a sample of blood was taken from each mouse and the forty (40) samples were subjected to duplicate analyses for hemoglobin.

- (a) Make a schematic representation of this experimental plan.
(b) Complete the following analysis of variance and write out the expected values of the mean squares:

| Source of Variation | df | Sum of Squares | M.S. | B.M.S. |
|--------------------------------------|----|----------------|------|--------|
| Total | 79 | 5760 | | |
| Diets | | 1800 | | |
| Mice within Diets | | | 100 | |
| Analyses within Mice
within Diets | | | | |

- (c) Test the significance of the effect of the diets on the hemoglobin content of the blood and make a probability statement of interpretation.

Master's Examination
Part II

- (d) Which is the greater source of experimental variation -- Rice or Analyses?
- (e) What magnitude of difference in the weight gain content of two of the dietary means would be needed to be considered significant at the 1% probability level?

6. In order to characterize the effect of food consumption (X) on the gain in weight (Y) of animals on a certain diet twenty (20) rats were individually caged and their grams of feed consumed and grams of weight gains were measured for the interval of interest. The following calculations have been made:

| | |
|-------------------|--|
| <u>Sums</u> | <u>Corrected Sums of Squares, etc.</u> |
| $\Sigma Y = 1200$ | $\Sigma Y^2 = 600$ |
| | $\Sigma XY = 180$ |
| $\Sigma X = 400$ | $\Sigma X^2 = 90$ |

- (a) Is there any correlation between feed consumption and weight gain? If so express it quantitatively.
- (b) Complete the regression analysis, testing the significance of the regression of gain on feed consumption.
- (c) Compute the regression equation for the prediction of weight gain from food consumption.
- (d) What are the 95% probability limits for the prediction (at the mean food consumption for this study) of weight gains from food consumption?
- (e) Construct a 95% confidence interval and make the appropriate verbal statement for the regression coefficient.

INTEGRAL AND SPACIAL DIFFERENTIALS

Wissenschaften, Geographie und Wirtschaftswissenschaften

Seite 1

Datum: 12.12.2020

Aufgabe 4 (aus der 1. Klausur, 2019/2020, 2020/2021, 2021/2022)

Aufgabe 4 (aus der 1. Klausur, 2019/2020, 2020/2021, 2021/2022)

Sei $f: \mathbb{R}^n \rightarrow \mathbb{R}$ eine Funktion, die in einem Punkt $x_0 \in \mathbb{R}^n$ differenzierbar ist.

$$Df(x_0) = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$$

(a) Berechnen Sie die Richtungsableitung von f in x_0 in Richtung des Vektors $v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

(b) Berechnen Sie die Richtungsableitung von f in x_0 in Richtung des Vektors $v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

(c) Berechnen Sie die Richtungsableitung von f in x_0 in Richtung des Vektors $v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

Sei $f: \mathbb{R}^n \rightarrow \mathbb{R}$ eine Funktion, die in einem Punkt $x_0 \in \mathbb{R}^n$ differenzierbar ist.

(a) Berechnen Sie die Richtungsableitung von f in x_0 in Richtung des Vektors $v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

(b) Berechnen Sie die Richtungsableitung von f in x_0 in Richtung des Vektors $v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

(c) Berechnen Sie die Richtungsableitung von f in x_0 in Richtung des Vektors $v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

(d) Berechnen Sie die Richtungsableitung von f in x_0 in Richtung des Vektors $v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

(e) Berechnen Sie die Richtungsableitung von f in x_0 in Richtung des Vektors $v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

Sei $f: \mathbb{R}^n \rightarrow \mathbb{R}$ eine Funktion, die in einem Punkt $x_0 \in \mathbb{R}^n$ differenzierbar ist.

(a) Berechnen Sie die Richtungsableitung von f in x_0 in Richtung des Vektors $v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

Chlorine, Chlorine gas,
 $\text{Cl}_2, \text{Cl}_2(g)$

1.0

(a) Chlorine gas can be obtained from HCl

(b) Chlorine gas can be obtained from HCl by electrolysis
 solution of HCl in water and H_2 and Cl_2

2. Chlorine gas can be obtained from HCl by electrolysis of HCl solution in H_2 and Cl_2

3. Chlorine gas can be obtained from HCl by electrolysis of HCl solution in H_2 and Cl_2

4. Chlorine gas can be obtained from HCl by electrolysis of HCl solution in H_2 and Cl_2

5. Chlorine gas can be obtained from HCl by electrolysis of HCl solution in H_2 and Cl_2

6. Chlorine gas can be obtained from HCl by electrolysis of HCl solution in H_2 and Cl_2

(a) Chlorine gas can be obtained from HCl by electrolysis of HCl solution in H_2 and Cl_2

(b) Chlorine gas can be obtained from HCl by electrolysis of HCl solution in H_2 and Cl_2

7. Chlorine gas can be obtained from HCl by electrolysis of HCl solution in H_2 and Cl_2

8. Chlorine gas can be obtained from HCl by electrolysis of HCl solution in H_2 and Cl_2

(a) Chlorine gas can be obtained from HCl by electrolysis of HCl solution in H_2 and Cl_2

(b) Chlorine gas can be obtained from HCl by electrolysis of HCl solution in H_2 and Cl_2

(c) Chlorine gas can be obtained from HCl by electrolysis of HCl solution in H_2 and Cl_2

(d) Chlorine gas can be obtained from HCl by electrolysis of HCl solution in H_2 and Cl_2

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Hauptkriterium für die Beurteilung
 der Qualität

erforderlich. (1) wird davon, dass bei Anwendung eines auf die beschriebene Weise

Wiederholung

- (a) demnach die in der ersten Phase
- (b) demnach die in der zweiten Phase

Wiederholung

- (c) demnach die in der dritten Phase

Wiederholung

Wiederholung

Wiederholung

(wiederholung)

10^{-10}

Wiederholung

Wiederholung

3. An der ersten Phase der Untersuchung

Wiederholung

Wiederholung

Wiederholung

Wiederholung

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1. The following are the results of the experiment:

Experiment 1: The results of the experiment are as follows:

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1. The results of the experiment are as follows:

4. (a) A fair die is tossed and a number is noted. Is there any way to tell the number of the die that was tossed? (b) A fair die is tossed and a number is noted. Is there any way to tell the number of the die that was tossed?

(c) A fair die is tossed and a number is noted. Is there any way to tell the number of the die that was tossed?

5. Suppose a fair die is tossed and a number is noted. Is there any way to tell the number of the die that was tossed?

6. Suppose a fair die is tossed and a number is noted. Is there any way to tell the number of the die that was tossed?

7. Suppose a fair die is tossed and a number is noted. Is there any way to tell the number of the die that was tossed?

8. Suppose a fair die is tossed and a number is noted. Is there any way to tell the number of the die that was tossed?

MEMORANDUM FOR THE RECORD

Washington, D.C.

December 30, 1949

Part II

Answer 1, out of 6 questions - Ben Book and Ben Baker

See also line book 1, 1949-50

1. Values of $\bar{I}$ and $\bar{J}$ have been located to be $\bar{I} = 1.597$ and $\bar{J} = 1.597$. The values of $\bar{I}$ and $\bar{J}$ are approximately 1.597, and are plotted on below. The values of $\bar{I}$ and $\bar{J}$ are plotted on page 2.

(a) Present the values of $\bar{I}$ and $\bar{J}$ as a function of $\bar{I}$ and $\bar{J}$ as a function of $\bar{I}$ and $\bar{J}$.

(b) Do you have a formula for the values of $\bar{I}$ and $\bar{J}$ as a function of $\bar{I}$ and $\bar{J}$?

(c) If the values of $\bar{I}$ and $\bar{J}$ are plotted on page 2, do you have a formula for the values of $\bar{I}$ and $\bar{J}$ as a function of $\bar{I}$ and $\bar{J}$?



Section 1001, Sec. 1001
Subj. 10

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- (a) It shall be the duty of the Secretary to determine and report to the President the results of the investigation of the activities of the persons named in the report of the President.
- (b) The Secretary shall determine the results of the investigation of the activities of the persons named in the report of the President.
- (c) The Secretary shall determine the results of the investigation of the activities of the persons named in the report of the President.
- (d) The Secretary shall determine the results of the investigation of the activities of the persons named in the report of the President.
- (e) The Secretary shall determine the results of the investigation of the activities of the persons named in the report of the President.

Michigan State University

Mathematics Final Examination

March 15, 1969
8:30-10:00 A.M.

Part I - Theory

Open Book Exam

Examinees will be picked up promptly at 10:00 A.M.

Answer all questions.

1. A die is thrown as long as necessary for an ace to come up. Supposing the number of throws n is odd, what is the probability $p = ?$

$$2. f_0(x) = (2\pi)^{-\frac{1}{2}} e^{-\frac{1}{2}x^2}$$

and

$$f_1(x) = (2\pi)^{-\frac{1}{2}} e^{-\frac{1}{2}(x-1)^2}$$

Let $f = f_0$ be the null hypothesis and $f = f_1$ the alternative. A critical region, $1/2 \leq x \leq 3/2$, is selected and a single observation taken.

- What is the Type I error?
- What is the Type II error?
- What is the power of the test?
- Is this the best one-sided critical region of this size? Prove or disprove.
- If the answer to (d) is "Yes" prove your statement. If the answer is "No" give an alternative critical region and state in what sense it is better.

3. If $X_1, X_2, \dots, X_n$ is a random sample of size n from a normal distribution with mean μ and variance σ^2 , find the best invariant unbiased estimator of σ^2 when μ is unknown. $\frac{1}{n} \sum_{i=1}^n (C_i X_i - A)^2$ where C_i and A are constants, $C_i \neq 0$. Prove that $\frac{1}{n} \sum_{i=1}^n (C_i X_i - A)^2$ is normally distributed.

4. (a) For a random sample of size n find the maximum-likelihood estimator for the parameter of the Poisson distribution.
- (b) Is it sufficient?
- (c) Is it unbiased?
- (d) Is it minimum variance?

Prove your statements.

Wally and the Duke University

Historic Final Examination

March 15, 1969
10:30 - 12 noon

Part II - Applied

Open Book Exam

Papers will be picked up promptly at 12 noon.

Answer all questions.

- Suppose we have just completed the following field experiment in the hope of recommending a "best" corn to New Jersey farmers. Three geographical locations and three varieties of corn were included in the investigation. The response was bushels per acre

| Location | Corn | | | Totals |
|----------|-------|-------|-------|--------|
| | A | B | C | |
| North | 20.24 | 14.13 | 12.8 | 100 |
| Central | 17.16 | 12.10 | 36.33 | 100 |
| South | 15.18 | 23.29 | 34.17 | 127 |
| Totals | 107 | 107 | 123 | 336 |

- Complete the ANOVA given that

TSS = 1008

S.S. for residual (error) = 74

S.S. for corn = 14.1

S.S. for location = 43.76

- Discuss the results in (a) and give your recommendations

- An experimenter in a drydock was interested in testing the effectiveness of 5 barracuda resistant paints. Four ships were chosen for the experiment. An experimental area on the hull of each ship was defined. The area was then divided into 5 sections and the 5 paints were then randomly assigned to a section. After 6 months in the water, the barracudas were scraped from 5 sample areas (1 square foot) within each of the 5 sections

- What is the objective of this experiment?

- Write out the ANOVA table and give the S.S.

- What is the conclusion of this experiment?

Parsons's Final Examination
Part II: Applied

3. It was generally felt that the length of a baby's foot and the length of its middle finger were correlated, but the degree of correlation was debatable. Hence a random sample of 107 babies was taken to make such an estimate. From the sample, $r = 0.48$ was calculated. If these lengths are distributed normally, what is a 95% confidence interval estimate of the degree of correlation in the population? State the meaning of this estimate in your own words.
4. Someone claimed that a person could not distinguish between two foods eaten when he could not smell, especially when the foods were about alike in texture and flavor. An expert selected two foods, A and B, as being types which would qualify under the claim if it were true. A test was arranged as follows:
- Each of 100 persons was given four portions at random to taste, three of one type and one of the other. All were nose clips during the test. Each was asked to pick out the "odd" portion.
- (a) State the null and alternative hypotheses.
- (b) Test the claim at the 5% level. If 30 of the 100 people correctly identified the "odd" portion.
- (c) What conclusion can be drawn from this experiment?

1. The first part of the document is a list of the names of the persons who have been appointed to the various positions of the Board of Directors of the Corporation.

2. The second part of the document is a list of the names of the persons who have been appointed to the various positions of the Board of Directors of the Corporation.

3. The third part of the document is a list of the names of the persons who have been appointed to the various positions of the Board of Directors of the Corporation.

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1. The first part of the paper is devoted to the study of the properties of the function $f(x)$ defined by the equation

$f(x) = \int_0^x \frac{1}{1+t^2} dt$

It is well known that this function is the arctangent function, i.e., $f(x) = \arctan x$. The properties of this function are well known, but we shall study them in detail for the sake of completeness.

2. In the second part of the paper, we shall study the properties of the function $F(x)$ defined by the equation

$$F(x) = \int_0^x \frac{1}{1+t^2} dt + \int_0^x \frac{1}{1+t^4} dt + \dots$$

It is easy to see that this function is increasing and concave down. We shall study its properties in detail.

3. In the third part of the paper, we shall study the properties of the function $G(x)$ defined by the equation

$$G(x) = \int_0^x \frac{1}{1+t^2} dt + \int_0^x \frac{1}{1+t^4} dt + \dots + \int_0^x \frac{1}{1+t^{2n}} dt + \dots$$

It is easy to see that this function is increasing and concave down. We shall study its properties in detail.

1. The first part of the paper is devoted to the study of the properties of the function $f(x)$ defined by the equation

$$f(x) = \int_0^x \frac{1}{1+t^2} dt, \quad x \in \mathbb{R}.$$

It is shown that the function $f(x)$ is increasing and concave down on the interval $(-\infty, \infty)$. Moreover, the function $f(x)$ is bounded on the interval $(-\infty, \infty)$.

2. The second part of the paper is devoted to the study of the properties of the function $g(x)$ defined by the equation

$$g(x) = \int_0^x \frac{1}{1+t^4} dt, \quad x \in \mathbb{R}.$$

It is shown that the function $g(x)$ is increasing and concave down on the interval $(-\infty, \infty)$. Moreover, the function $g(x)$ is bounded on the interval $(-\infty, \infty)$.

3. The third part of the paper is devoted to the study of the properties of the function $h(x)$ defined by the equation

$$h(x) = \int_0^x \frac{1}{1+t^6} dt, \quad x \in \mathbb{R}.$$

It is shown that the function $h(x)$ is increasing and concave down on the interval $(-\infty, \infty)$. Moreover, the function $h(x)$ is bounded on the interval $(-\infty, \infty)$.

4. The fourth part of the paper is devoted to the study of the properties of the function $k(x)$ defined by the equation

$$k(x) = \int_0^x \frac{1}{1+t^8} dt, \quad x \in \mathbb{R}.$$

It is shown that the function $k(x)$ is increasing and concave down on the interval $(-\infty, \infty)$. Moreover, the function $k(x)$ is bounded on the interval $(-\infty, \infty)$.

5. The fifth part of the paper is devoted to the study of the properties of the function $l(x)$ defined by the equation

$$l(x) = \int_0^x \frac{1}{1+t^{10}} dt, \quad x \in \mathbb{R}.$$

It is shown that the function $l(x)$ is increasing and concave down on the interval $(-\infty, \infty)$. Moreover, the function $l(x)$ is bounded on the interval $(-\infty, \infty)$.

1. The first part of the paper discusses the importance of the study of the history of the United States. It is argued that a knowledge of the past is essential for a full understanding of the present and for the development of a sound policy for the future.

2. The second part of the paper discusses the role of the government in the development of the United States. It is argued that the government has played a crucial role in the development of the country, and that its actions have been guided by a set of principles that have been passed down from generation to generation.

3. The third part of the paper discusses the role of the individual in the development of the United States. It is argued that the individual has played a crucial role in the development of the country, and that his actions have been guided by a set of principles that have been passed down from generation to generation.

4. The fourth part of the paper discusses the role of the future in the development of the United States. It is argued that the future is a time of great opportunity, and that it is up to us to make the most of it. It is also argued that the future is a time of great challenge, and that it is up to us to meet the challenges that lie ahead.

